



PLACEMENTS THRU' INTERNSHIPS

2026-27

OFFER DOCUMENT

Empowering Students

for

Full Stack Technology Development

Thru'

Industry Oriented Internship Program

(Applicable to Final Year 2026-27 & Graduates 2025-26)

Open-Source – Open API - Multi-Domain – Cross-Industry)

AI Services Enabled, Digital Process Platform.



June 2026



smart-KYC
AI Services



Alpha FinSoft Pvt. Ltd.

9th June 2026

To:

The Principal / Placement officer / Internship Students.

Subject:

- Placements thru' Internships 2025-26
- Invitation to Collaborate on Industry-Integrated Placements thru' Internship Program in "Regulatory Technology" (RegTech) Platforms for CS/IT Students
- **Open to Recent Graduates 2025.**

Dear Sir/ Madam,

I hope this message finds you well.

We are pleased to extend an exclusive internship opportunity to the students of your esteemed institution, aimed at nurturing practical engineering talent by exposing students to real-world, industry-relevant platforms in the following techno-functional areas in BFSI Domain sector:

About the Internships:

BFSI Domain Areas:

- Open KYC, Open Document Management System (Open DMS), and Open API Gateway—a suite of platforms aligned with Regulatory Technology (RegTech) practices adopted by leading institutions such as RBI, SEBI, IRDAI, and under compliance mandates like PMLA.

Broader Purpose:

- This initiative is part of our broader strategy to promote the Open Work flow Ecosystem through industry-academia collaboration and hands-on project-based learning for future-ready technologists.

Internship Offer:

- **The Internship Offer is open to recent graduates of (B.E. - C.S, IT; B.Sc. C.S., IT); (B.E., C.S, IT) &**
- **Internship Period, 3 Months, Full-Time (for Graduates), or Part-Time (On-site).**

Internship Modes:

- All the Internship Programs are On-Site with technical support from Sr. Members of Alpha FinSoft Pvt. Ltd.
- Full-time Employment is offered, post internships based on the performance.



About the Products & Platforms – Alpha FinSoft Pvt. Ltd.

- ◆ 1. Open API Digital KYC Platform



Alpha FinSoft Pvt. Ltd.

A digital, modular, and secure Know-Your-Customer platform designed for Banking and Financial Services, facilitating seamless KYC lifecycle management—capture, verification, storage, and renewal—while staying compliant with regulatory norms.

Key Functionalities:

- Multi-product KYC onboarding
- API-driven verification
- Wallet and transaction-based risk profiling
- AI-powered document and facial recognition
- Integration with UIDAI, PAN, CKYC, and DigiLocker

Architecture Highlights:

- Microservices-based backend using .NET Core
- API Gateway exposing all services via OpenAPI (Swagger)
- Modern React-based frontend with reusable components
- AI/ML utilities for document classification and fraud detection
- PostgreSQL for data storage, secured with encryption
- NGINX as a web server and load balancer

◆ 2. Open API Document Management System (DMS) Platform

An open, secure, and AI-powered Document Management System for enterprises to digitize, tag, retrieve, and archive documents.

Features:

- OCR-based scanning & indexing
- AI-assisted tagging and retrieval
- Workflow-based document approval
- Audit trails & user access control
- Custom folder hierarchies and cloud integration

Architecture:

- REST-based microservices for document operations
- Blob storage + PostgreSQL metadata
- Integration-ready APIs for third-party apps

◆ 3. Open API Gateway Platform

A scalable, lightweight API management layer that exposes backend services through a unified, secure, and discoverable interface.

Features:

- Rate limiting, authentication, and version control
- OpenAPI specs with test and monitor tools
- Role-based access controls
- Real-time API analytics and performance tracking
- AI Enhancements:
- Smart request routing using ML-based predictions
- Auto-documentation generation and semantic analysis



Internship Structure and Opportunities



We are inviting applications for a limited batch of 40 students (20 per Year) from 3rd and 4th year B.E. (CS/IT) and final year B.Sc. (IT) programs & Post Degree completed Students of both B.E. & B.Sc.

◆ **Objectives:**

- Learn and apply modern software architecture concepts
- Understand compliance-centric application development
- Build functional POCs (Proofs of Concept) for BFSI use cases
- Showcase AI-enabled enhancements in real-world APIs and platforms
- Collaborate with professionals and partner ecosystem for mentorship

Page 3 of 5

◆ **Internship Details:**

- **Existing Students:** Duration: 6 + 6 Calendar months (semester-based).
- **Recently Graduated Students-** 3 months Full Time – On-Site with customer project allocation in the 1st month.
- Support Frequency from Alpha FinSoft Pvt. Ltd: One session per week (On-line Briefing Sessions on Saturday, up to 2 hrs, by our Sr. Project Teams)
- Mode: Hybrid – college lab or personal Laptop or Desktop based (Pl. check required configuration)
- Batch Size: 10 students per college per semester

Tools Covered:

- Backend: .NET Core, RESTful API design
- Frontend: React, UI/UX best practices
- Database: PostgreSQL
- Web: NGINX, OpenAPI Gateway
- AI Tools: OCR, NLP, ML for verification and classification

Pre-Requisites: (Structured Technology Learning Path, separately enclosed)

- Basic knowledge of programming (C#, JavaScript), as per the Learning Path, separately enclosed.
- Understanding of web development
- Willingness for self-learning and self-paced development
- Personal laptop/desktop with at least:
- 8GB RAM (Preferred 16 GB), i5 processor or higher
- Internet connectivity and Visual Studio Code/IDE

(Self-Learning, on-line material will be provided. Prior experience is NOT mandatory.)



Why This Internship? – Benefits to Students

- Practical application of academic concepts in a real-world BFSI/RegTech domain
- Mentored POC development that can be showcased during placements
- Networking opportunities with professionals and industry mentors
- Certification & recommendation letters
- Top performers may be offered placement or project opportunities in our partner ecosystem



Career Relevance

This internship will position students to take on roles such as:

- RegTech Engineer
- API Developer
- Platform Architect
- AI & Data Analyst
- Full Stack Developer
- Compliance Product Specialist



Their exposure to AI-driven digital platforms will give them a competitive advantage in placement drives, entrepreneurship, or higher studies.



Next Steps

- We propose an introductory session/demo at your campus to showcase the platforms
- Interested colleges may nominate up to 10 eligible students for this internship
- Confirm participation by replying to this letter or contacting our representative

We look forward to partnering with your esteemed institution in nurturing the next generation of impactful tech talent. Page 4 of 5

Warm Regards,

For Alpha FinSoft Pvt. Ltd.

Aishwarya Patil
HR & Administration

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Attachments:

- INDO-FinNET - Digital KYC Products' Literature
- Platform Architecture Diagrams (Open KYC, DMS, API Gateway)
- Structured Internship Program
- Internship Learning Paths Links (Self-Learning)
Key Technologies - .NET Core, React, PostgreSQL
- **Internships Registration Form – On-line Registration Link**
<https://forms.gle/nwC8ncqpEXYjsXR9A>
- **Internship Information Documents Download Link**
<https://drive.google.com/drive/folders/1eEyAqWml8xcDwkeorT3jIXrgjfOvIOHY?usp=sharing>



Streamline Your Physical KYC to Digital KYC!

Integrated Platform for Digital KYC, Customer On-Boarding & Paperless Account Opening.

Digital KYC Methods

- Aadhaar – On-Line & XML
- PAN, DL, PP, Voter ID Verification
- C-KYC, Search & Download
- Live Photo with LA-LO, Date-Time Stamp
- Video KYC – Two & One Way, Call Center Queuing Support.
- Digital KYC Doc. Mgt.
- DigiLocker Services
- Bank A/C Name Verification or Penny Drop Services
- Address Look-up & Verification Services

Digital Signatures

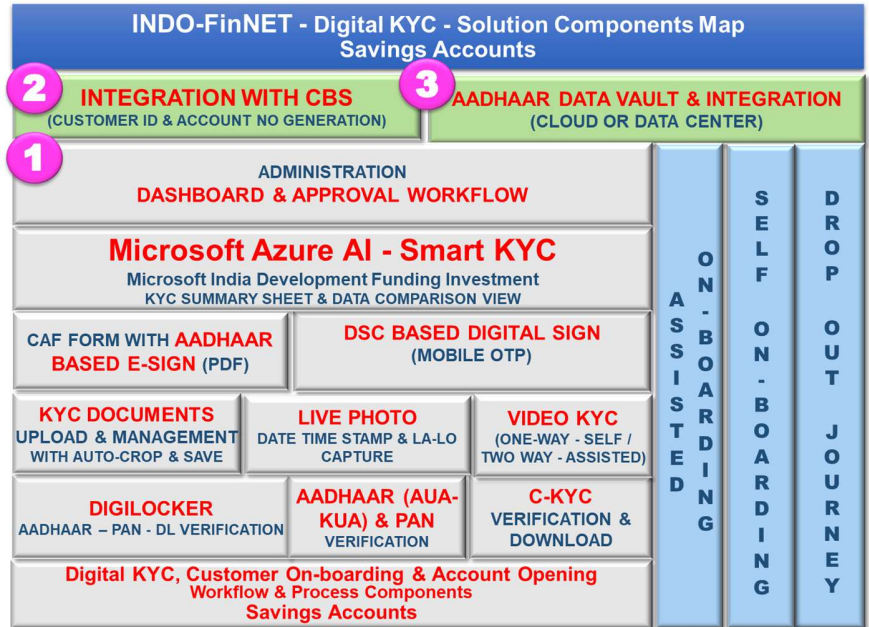
- Aadhaar E-Signs
- Digital Mobile Sign

Regulatory Compliance

- RBI, IRDA, SEBI, UIDAI, PMLA, DPDP Act 2023.
- Aadhaar Security - Data Vault & Aadhaar Masking

About INDO-FinNET

- Digital Customer On-Boarding Platform with Paperless KYC & Account Opening.
- KYC E-Document Management
- License Platform or Cloud ASP Subscription Model.
- Microsoft Azure AI Intelligence into KYC Process.



INDO-FinNET Features

On-Boarding Journey:

- Customer Self On-Boarding.
- Agent or RM based, Assisted On-Boarding
- Drop-Out & Missing KYC Journey Support.
- Maker-Checker based Approval & Rejection Flows
- Centralized or Decentralized Processing Support.

Regulatory Features:

- Digital Signs Support - Aadhaar E-Sign or Mobile OTP.
- Mobile, E-Mail validations.
- Live Photo, LA-LO & Date-Time Stamp Capture.

Process Automation Features:

- KYC Comparison Summary Sheet.
- Integration support with Core Business System.
- Microsoft AI Power – Image Recognition & Extraction, Data Extraction, Signature & Photo Extraction, Liveliness Factor.
- MIS & Admin Module with Approval Workflow.

Prominent Live Customers:

- Saraswat Bank, Cosmos Bank, AB FSL, AB HFC, Birla Sun-Life, Star Union Dai-Ichi Life Insurance & more...

Business Partnerships & Certifications



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Digital KYC & C-KYC Compliance - Unified KYC Process Automation

I. Executive Summary

“INDO-FinNET 2.0-Global” Customer Identification Platform is an AI-powered, In-Premise or cloud-ready, Digital KYC and On-boarding Solution.

The Platform enables financial institutions to verify customer identity through Digital Services, onboard users either “In-Presence” or “Remotely” (Video KYC) and open accounts seamlessly—without paper, delays or branch visits.

This document provides techno-functional approach for adopting a unified Digital KYC and C-KYC Automation in alignment with the RBI's KYC Master Direction (November 6, 2024).

RBI's New KYC Master Direction & CERSAI Regulatory Directives

The Reserve Bank of India (RBI), through its amended KYC Master Direction dated November 6, 2024, mandates:

- i. Pre-account-opening C-KYC Search & Download using the customer's C-KYC Identifier (KI).
- ii. Update to CKYCR within 7 days of any Customer Due Diligence (CDD).
- iii. Reverse updates into CBS/LMS upon changes notified via unsolicited responses from CKYCR.

This necessitates full automation of C-KYC processes with seamless integration to Digital KYC platforms to maintain regulatory compliance, minimize rejections, and ensure data fidelity.

It outlines the regulatory requirements, highlights key operational and financial benefits, compares deployment models (in-premises vs cloud), and proposes a scalable SaaS-based solution that small and large banks can adopt in making KYC fully compliant with regulatory guidelines.

INDO-FinNET 2.0, Digital KYC Platform fully complies with RBI's Digital KYC Master Direction.

Further its integrated C-KYC Automation Approach for Account Opening Process makes manual KYC Documents processing completely redundant.

Digitally Transform Your KYC Process with INDO-FinNET 2.0

Compliant, Cost-Effective & Complete

Many financial institutions in India still rely solely on Aadhaar OTP for KYC, but this method no longer meets the requirements of Full KYC as per RBI and PMLA regulations.

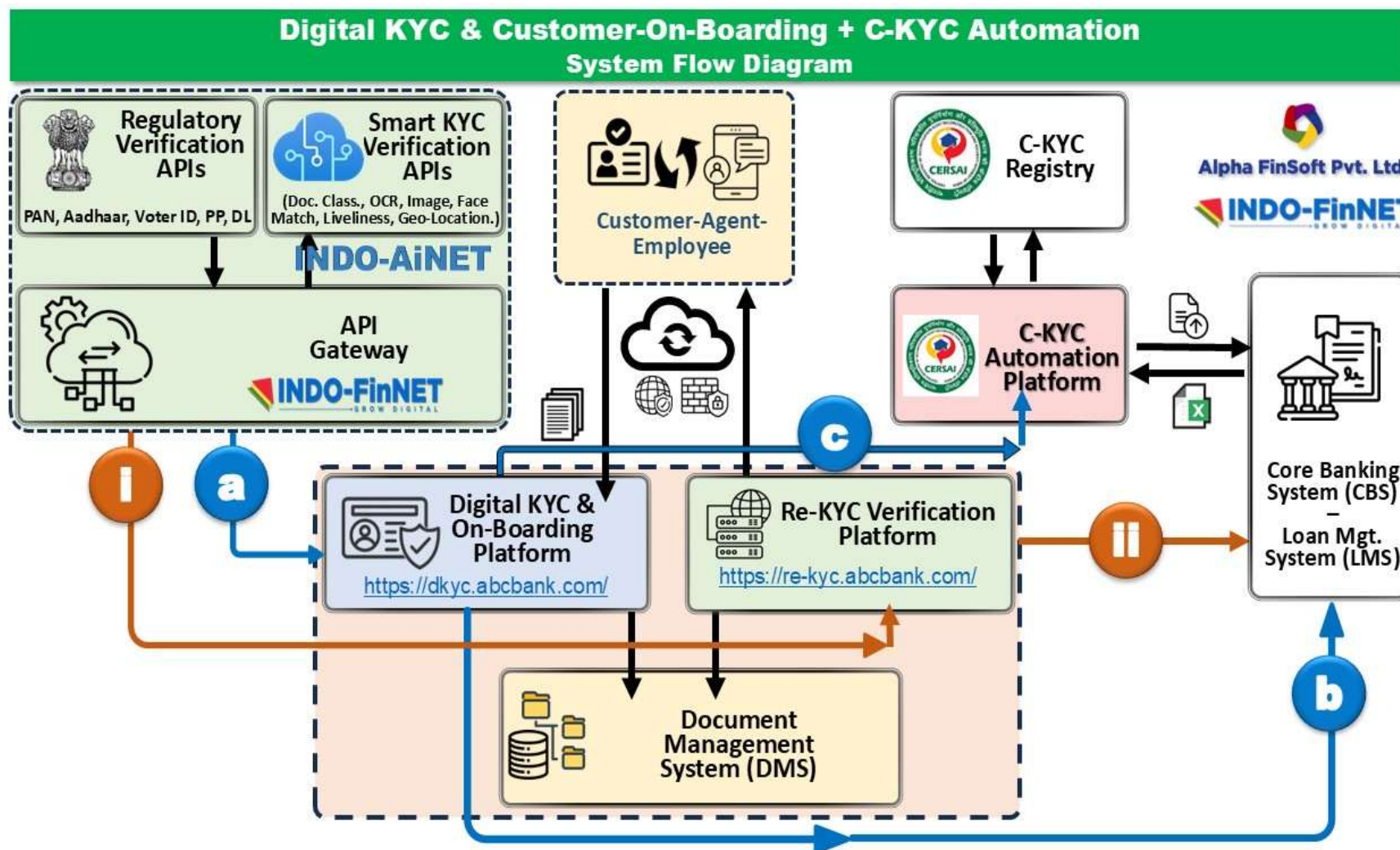
INDO-FinNET 2.0 by Alpha FinSoft offers a comprehensive digital KYC platform that goes beyond OTP.

It supports both Aadhaar and non-Aadhaar identity documents, automates C-KYC, enables Aadhaar – Biometric (AUA-KUA or Sub-AUA-KUA) or DigiLocker API integration without the requirements of DigiLocker App registration by the customers, and includes features like live photo capture and In-Person Verification (IPV).

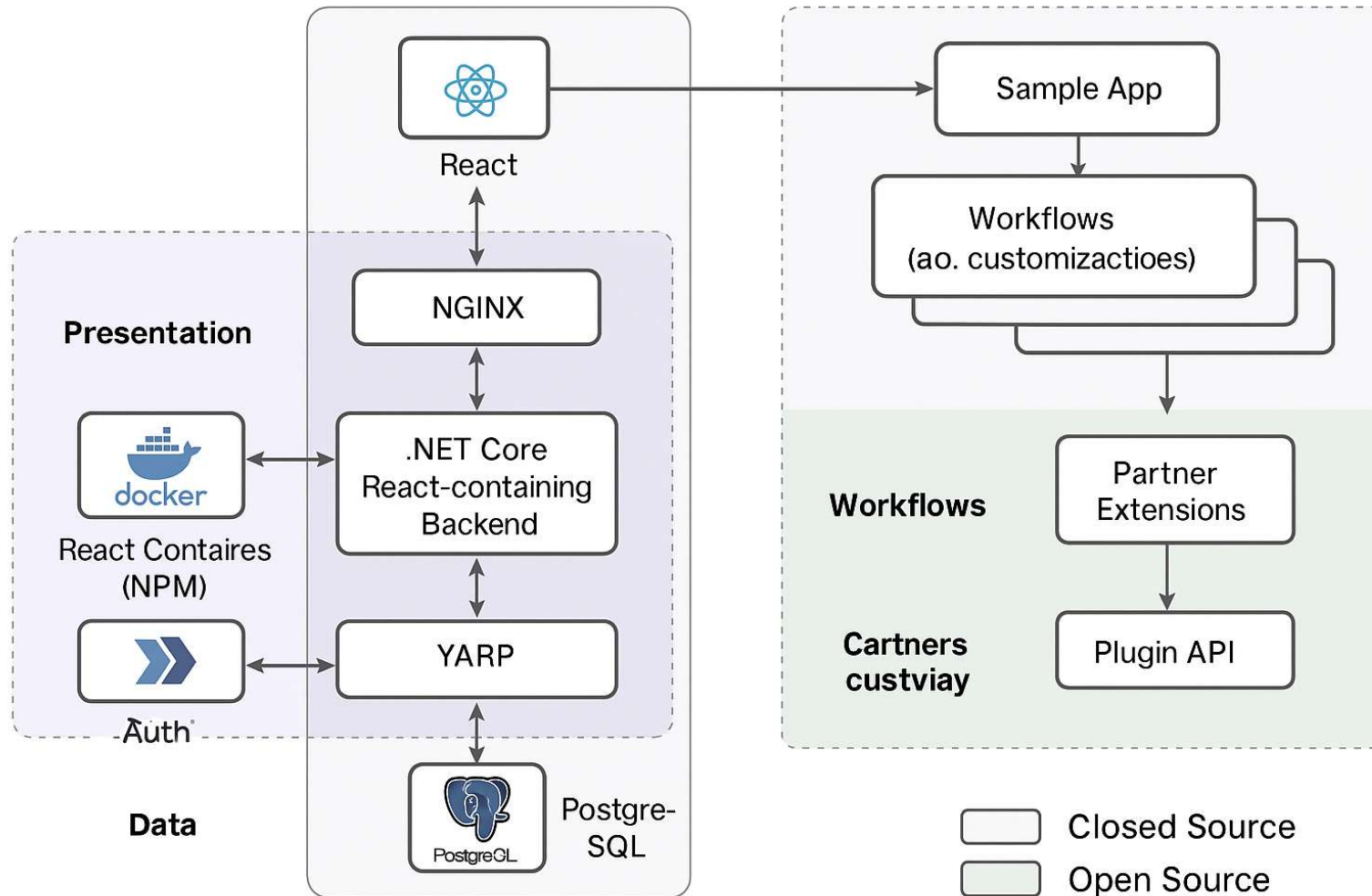
Unlike Aadhaar Biometric systems (viz. Aadhaar AUA-KUA or Sub-AUA-KUA), it eliminates the need for costly UIDAI certifications—making it ideal for banks, NBFCs, and credit societies.

The Platform facilitates to stay legally compliant, reduce infrastructure costs, and avoid non-compliant practices like screen scraping Aadhaar XML data.

Adopt INDO-FinNET 2.0 and upgrade to a secure, scalable, and regulation-ready KYC system.



OPEN API- DIGITAL PROCESS & WORKFLOW PLATFORM



Alpha FinSoft Pvt. Ltd. - Structured Internship Program



Program Overview

Alpha FinSoft Pvt. Ltd. offers a structured, semester-wise internship for final-year students of **BE/BTech** and **BSc (IT/Computer Science)** to build real-world experience in software engineering, microservices, compliance systems, and modern UI workflows. This program provides hands-on learning with internal and live customer projects while fostering mentorship and team collaboration.



Semester-wise Internship Structure (Interns are required to have Own Laptop)

Semester	Applicable To	Duration	Time Commitment	Internship Certificate	Stipend	Activities Summary
5th & 6th	• BE/BTech, BSc (IT/CS)	6 months/ Sem	• 1 hr/day (Weekdays) + Sat 2 hrs	Yes	No	• Free Self-Learning Paths + Tech POCs
7th	• BE/BTech only	6 months	• 2 hrs/day (Mon–Fri) + Sat 4–6 hrs	Yes	No	• Internal Projects + Crash Learning + POCs (Assignments)
8th	• BE/BTech only	6 months	• 2 hrs/day (Mon–Fri) + Sat 4-6 hrs	Yes	₹500 (Top 5 only)	• Live Customer Projects + Client Interactions
Recent Graduates	• All above	3 months	• 8 hrs/Day	Yes + Job Offer OR Placement Assistance	No	• All the above Stages

Max Intake: 15 students per semester per college



Learning Areas & POC Topics

Category	Focus Areas / Sample Projects
• Government APIs	• Aadhaar eKYC, PAN, GST, Digilocker, GST, E-Sign integrations
• Functional APIs	• Wallets, Account Opening, Loans, KYC Flows, Threshold Engines, Video KYC
• Technology Services	• .NET Core Microservices, Swagger APIs, PostgreSQL
• Architecture	• Microservices, Containers, Docker, NGINX, CI/CD, Multitenancy
• UI + Workflow	• React, Form Builder, Validation Engine, Workflow Renderer
• Industry Workflows	• Banking (Current/Loan), ISO Process Automation, HR Onboarding



Technology Stack & Tools

- **Backend:** .NET Core Web API, PostgreSQL
- **Frontend:** React, Tailwind CSS, Storybook
- **DevOps:** Docker, GitHub Actions, NGINX
- **Documentation:** OpenAPI (Swagger), Markdown, ERD Tools
- **Communication:** Google Meet, WhatsApp, Email
- **Versioning:** GitHub (Private Repos)



Intern Expectations & Model

- **Daily Learning:** Minimum 1 hour (5th/6th Sem); 2 hours (7th/8th Sem)
- **Weekly Sync:** Saturday briefings and query resolution (On-line)
- **Deliverables:** POCs, Assignments, learning logs, GitHub submissions
- **Presentation:** One per intern per semester
- **Mentorship:** Seniors to mentor juniors; peer learning encouraged and will be counted as one of the criteria for absorption.



Collaborative Learning Clause

- Interns are expected to **mentor junior students** through peer learning.
- Final project **presentations are mandatory**.
- Colleges are expected to host review and knowledge-sharing sessions.



Terms & Conditions

1. **Ownership of Work:**
All source code, POCs, documentation, and designs developed during the internship shall be the **exclusive intellectual property of Alpha FinSoft Pvt. Ltd.**
2. **Non-Monetary Periods:**
No stipend is applicable during the 5th, 6th, and 7th semesters. A **stipend of ₹2,000/month** may be payable to **top 5 students in the 8th semester** based on project performance and internal requirements of Alpha FinSoft Pvt. Ltd.

No Stipend is payable to Full-Time (3 months) Interns.
3. **Certificates:**
Internship certificates will be awarded upon successful completion, including:
 - Attendance
 - Project submission
 - Final presentation
4. **Device & Infrastructure:**
 - Interns must **carry their own laptops** throughout the internship.
 - Alternatively, colleges must provide **labs with required software and internet access**.
 - Alpha FinSoft Pvt. Ltd. is not responsible for providing hardware or infrastructure.
5. **Job Opportunity Post-Internship:**
 - Up to **5 interns may be offered employment** based on merit and project output.
 - Students are **free to seek opportunities elsewhere**, if no offer is made by Alpha FinSoft Pvt. Ltd.
6. **Confidentiality:**
Interns must maintain **confidentiality** and avoid sharing any internal or client data publicly. Violation will result in termination of internship.
7. **Conduct & Participation:**
Active participation, discipline, and timely task delivery are mandatory for internship continuation.
8. **Selection Limit:**

Only **15 students per semester** will be selected based on screening by Alpha FinSoft Pvt. Ltd.



Post Internship Absorption Strategy

- Monthly interviews for top-performing interns
 - Conversion into full-time employment
 - Fast-track roles for those with client exposure and strong communication
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**Expectations from Colleges**

- Official nomination and consent from departments
 - Hosting student presentations and peer-learning events
 - Providing attendance confirmation and academic compliance
 - Lab infrastructure if students lack personal devices
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Technology Stack Learning Path

1. .NET Core Learning Path

Here's a structured **.NET Core Developer Learning Path** tailored for your needs—especially for building scalable microservices for the **Open Digital KYC platform**.

It includes resources, videos, docs, and a week-wise plan from fundamentals to architecture-level proficiency.

✓ .NET Core Developer Learning Path (8–10 weeks)

Week	Topics	Goals	Resources
Week 1	• .NET & C# Fundamentals	• Understand C# syntax, OOP, solution structure, CLI	- C# Fundamentals – Microsoft Learn - C# Full Course (YouTube) - .NET CLI – MS Docs
Week 1	• ASP.NET Core Basics	• Build basic web APIs, dependency injection	- Build First Web API – MS Learn - ASP.NET Core for Beginners (YouTube)
Week 1	• Entity Framework Core (EF Core)	• Work with DbContext, migrations, LINQ	- EF Core Docs - EF Core Crash Course (YouTube)
Week 1	• RESTful API Design	• Build modular services, Swagger, versioning	- Designing RESTful APIs - Use Swagger in .NET Core
Week 2	• Middleware, Filters, Logging	• Understand pipeline, global error handling	- Middleware in ASP.NET Core - Logging Providers
Week 2	• Authentication & Authorization	• JWT, Identity, Role-based access	- Authentication in ASP.NET Core - JWT Auth in Web API (YouTube)
Week 2	• Microservices & Clean Architecture	• Services, Interfaces, Dependency Injection	- Clean Architecture in .NET - Microservices Architecture Guide (MS)
Week 2	• Async Programming & Messaging	• Async/await, Task, RabbitMQ/Kafka basics	- Async in C# - Messaging Patterns
Week 2	• Docker & Deployment	• Dockerize your .NET app, local containers	- Containerize .NET App - Docker for .NET Devs (YouTube)
Week 2	• API Gateway & Observability	• Integrate Ocelot API Gateway, metrics, tracing	- Ocelot Gateway Docs - Distributed Tracing with OpenTelemetry

1. Once you complete this .NET Core path, you'll be ready to:

- Design and implement scalable APIs for the **Digital KYC platform**
- Deploy microservices with **Docker + API Gateway**
- Work confidently with **React UI + .NET Core backend**

2. REACT UI Developer Learning Path

Here's a step-by-step React developer learning path (focused beyond just UI), including the right reading material, video resources, and projects to help you gain expertise suitable for building and contributing to your Digital KYC platform.

Goal

Build full React proficiency — not just UI — but also state management, component architecture, API integration, testing, and deployment, aligned with enterprise-grade apps like your Digital KYC platform.

Phase 1: Fundamentals of React

What to Learn

- JSX
- Functional Components
- Props & State
- Basic Hooks: useState, useEffect

Reading

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Videos

- [Beginner React – freeCodeCamp \(5 Hours\)](#)

Outcome

You can build small apps like a TODO app, counter, etc.

Phase 2: Component Architecture & Best Practices

What to Learn

- Component Folder Structure (atomic design / feature-based)
- Lifting state up & prop drilling
- Composition vs inheritance
- Controlled vs uncontrolled components

Reading

-
-

Videos

- [Clean Code in React \(ThePrimeagen\)](#)

Outcome

You'll design reusable components & modular file structures, key for large apps like your Digital KYC.

Phase 3: Advanced React & State Management

What to Learn

- Context API

- Redux Toolkit (RTK) or Zustand
- Custom Hooks
- Memoization: React.memo, useMemo, useCallback

Reading

-
-

Videos

- [Redux Toolkit Crash Course – freeCodeCamp](#)
- [Zustand in 15 min](#)

Outcome

You'll manage global state, local component state, and optimize performance for scalable apps.

Phase 4: API Integration & Async Logic

What to Learn

- Fetching data via fetch, axios
- API Status handling (loading, error, success)
- React Query (TanStack Query)
- Forms and validation (React Hook Form / Formik + Yup)

Reading

-
-
-

Videos

- [React Query Crash Course](#)
- [React Hook Form Tutorial](#)

Outcome

You'll integrate your .NET Core APIs with error handling, pagination, caching, etc.

Phase 5: Routing, Authentication, Authorization

What to Learn

- React Router v6+
- Private Routes
- Role-based Access Control
- Token storage (HttpOnly cookie / localStorage)

Reading

-
-

Videos

- [React Router 6 Tutorial](#)
- [JWT Auth in React](#)

🎯 Outcome

You'll be able to protect routes and manage auth securely across components.

📖 Phase 6: Testing, Debugging & Performance

✅ What to Learn

- Testing with React Testing Library (RTL)
- Unit vs Integration testing
- Performance profiling
- DevTools

📖 Reading

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-

👥 Videos

- [RTL Tutorial](#)

🎯 Outcome

You'll write testable, stable, production-ready components.

📖 Phase 7: TypeScript with React (Optional but recommended)

✅ What to Learn

- Typing props, state, events
- Generics in components
- Custom hook types

📖 Reading

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👥 Videos

- [React + TypeScript Crash Course](#)

🎯 Outcome

Strongly typed, maintainable React codebase.

📖 Phase 8: Production Ready Practices

✅ What to Learn

- Component libraries (MUI, Tailwind, shadcn/ui)
- Lazy Loading, Code Splitting
- Error boundaries
- CI/CD (basic GitHub Actions / Vercel / Netlify)
- Environment variables

📖 Reading

-
- [TailwindCSS + React Guide](#)

Videos

- [React CI/CD Deployment on Vercel](#)

Outcome

You'll ship real, scalable apps that are maintainable, performant, and secure.

Capstone Projects (Build Along)

Project	Concepts Practiced
• Admin Dashboard (Role-based UI)	• Auth, Role-Routing, Component Layouts
• Digital KYC Simulation	• Form validation, API handling, workflow UI
• Wallet UI	• Transactions, state updates, data tables
• Micro-frontend (optional)	• Module federation / scalability

Suggested Tools & Libraries for Digital KYC App

Area	Library
• State Management	• Redux Toolkit or Zustand
• API Layer	• React Query
• Forms	• React Hook Form + Yup
• UI Styling	• TailwindCSS or MUI
• Testing	• React Testing Library
• Authentication	• JWT with Private Routing
• Deployment	• Vercel, Netlify, or Azure Static Web Apps

Suggested Self Learning Path

Phase	Task	Start Date	End Date	Status
• Phase 1 – Fundamentals	• Read Official React docs: Main Concepts	2025-08-01	2025-08-03	☐
•	• Watch freeCodeCamp Beginner React video	2025-08-04	2025-08-04	☐
•	• Build TODO app	2025-08-05	2025-08-07	☐
• Phase 2 – Component Architecture	• Read “Thinking in React”	2025-08-08	2025-08-09	☐
•	• Structure components using atomic/feature design	2025-08-10	2025-08-11	☐

(Continue similarly through Phases 3–8, then Capstone Projects.)

✓ **Step-by-Step Guide to Using**

1. Open a new Google Sheet and paste in the table above (continue through all phases).
 2. Add checkboxes in the “Status” column → Insert → Checkbox.
 3. Use date formatting for clarity and filtering.
 4. To visualize as a timeline or Gantt chart, apply a project-timeline template (Google’s built-in gallery or third-party) and integrate your task table [ClickUp+1Spreadsheet Point+1Sheetakispreadsheetclass.com+6Unito+6TeacherCast Educational Network+6Coefficient+1The Applicious Teacher+1](#).
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 How to Share as a Template

- Name your sheet something like React-Roadmap-Template.
 - Click Share, set “Anyone with the link” → Viewer [YouTube+8Sheetaki+8Info Inspired+8](#).
 - Copy the shared link, then replace the segment after the last slash (/edit?...) with /copy or /template/preview. Share that as your template link [chartexpo.com+3Conquer the Digital Empire+3Info Inspired+3](#).
 - When others open it, they’ll see a “Use Template” or “Make a copy” prompt.
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 Example Checklist Items per Project Phase

- Phase 1: Learn hooks; build small apps (counter, TODO).
 - Phase 2: Organize reusable components; ensure clear file folder structure.
 - Phase 3: Add Redux Toolkit or Zustand; build small state-managed app.
 - Phase 4: Connect React to .NET Core API; implement loading/error UI.
 - Phase 5: Secure auth/UI; implement private routes.
 - Phase 6: Write unit & integration tests with React Testing Library.
 - Phase 7: Convert components to TypeScript types.
 - Phase 8: Introduce code-splitting, lazy loading, deploy via Vercel/Azure.
 - Capstone: Build a mini-Digital KYC workflow with form flows, APIs, roles.
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3. PostgreSQL Database Learning Path

Here's a step-by-step learning path for PostgreSQL specifically tailored for freshers who are developing applications in .NET Core.

This guide focuses on building expertise in PostgreSQL from fundamentals to advanced database programming, administration, and integration with .NET Core using Npgsql.

Here's a step-by-step learning path for PostgreSQL specifically tailored for freshers who are developing applications in .NET Core.

This guide focuses on building expertise in PostgreSQL from fundamentals to advanced database programming, administration, and integration with .NET Core using Npgsql.

Learning Path for PostgreSQL (for .NET Core Developers)

 **Goal:** Build end-to-end expertise in PostgreSQL for use with .NET Core applications, including database design, queries, procedures, performance tuning, and integration via Npgsql.

Phase 1: PostgreSQL Fundamentals

Topic	Objective	Resource Link
What is PostgreSQL?	Understand the basics and ecosystem	Intro to PostgreSQL – DigitalOcean
Installation (Windows/Linux)	Install PostgreSQL locally or via Docker	Official Docs / Docker Setup
SQL Basics	Learn SELECT, INSERT, UPDATE, DELETE	SQL Tutorial – Mode
Data Types in PostgreSQL	Learn about text, int, jsonb, timestamp, etc.	PostgreSQL Data Types
Basic Constraints	Use PRIMARY KEY, UNIQUE, NOT NULL, FOREIGN KEY	Constraints Guide
PgAdmin Tool	Learn GUI tool to manage databases	PgAdmin Docs

Phase 2: Intermediate PostgreSQL + Database Design

Topic	Objective	Resource Link
Normalization & ER Modeling	Design normalized schema with relations	DB Design Basics – Vertabelo
Joins, Grouping, Aggregations	Use INNER JOIN, LEFT JOIN, GROUP BY, HAVING	SQL Joins – Visual Guide
Indexing & Performance	Use B-tree, hash indexes	PostgreSQL Index Guide
Views, Materialized Views	Abstract queries, cache results	Views vs Materialized Views
Transactions & Isolation	Use BEGIN, COMMIT, ROLLBACK, isolation levels	Transactions Guide

Phase 3: Advanced PostgreSQL Programming

Topic	Objective	Resource Link
Stored Procedures / Functions	Write PL/pgSQL functions and triggers	PL/pgSQL Tutorial
Triggers	Use BEFORE, AFTER triggers on tables	PostgreSQL Triggers
JSON/JSONB Columns	Store/query dynamic data	JSON Support
CTEs & Window Functions	Use WITH, OVER(), ROW_NUMBER()	Window Functions
Upserts & Concurrency	Use ON CONFLICT, RETURNING, and locks	Upsert Docs
PostgreSQL Extensions	Use extensions like uuid-oss, pgcrypto, etc.	Extension Usage

Phase 4: PostgreSQL with .NET Core

Topic	Objective	Resource Link
Npgsql Setup	Install and connect PostgreSQL with .NET	Npgsql Docs
EF Core with PostgreSQL	Use Entity Framework Core with PostgreSQL	EF Core + Npgsql
CRUD with EF Core	Scaffold models, write basic DB logic	CRUD Example
Migrations	Handle DB schema changes	EF Core Migrations
Dapper with PostgreSQL	Use lightweight ORM for performance	Dapper + PostgreSQL
Stored Proc from .NET	Call PL/pgSQL functions from .NET	Call Stored Proc from Npgsql

✓ Phase 5: Admin & Deployment Skills

Topic	Objective	Resource Link
Backup/Restore	Use pg_dump, pg_restore	Backup/Restore Guide
User Roles and Permissions	Role creation, grants, security	Role Management
Performance Monitoring	Use EXPLAIN, ANALYZE, pg_stat_activity	Performance Tuning
Logs & Configuration	Configure postgresql.conf, logging	PostgreSQL Logging
Hosting/PostgreSQL Cloud	Use cloud DB (e.g., AWS RDS, Supabase)	Supabase PostgreSQL
Security & SSL	Enable SSL, encrypted passwords	Security Best Practices

✓ Phase 6: Project Practice & Capstone

Project Type	Description
Task Manager	Create a multi-user task manager in .NET Core with PostgreSQL backend
Audit Trail System	Build a logging/audit system using triggers and JSONB for change tracking
Wallet or KYC Module	Implement transaction-based logic for Open KYC or Wallet APIs
API Performance Testing	Use Postgres indexing & tuning for APIs with EF/Dapper